

Fossil_Tablet error

Menghan Fu

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```
library(tibble)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(ggplot2)
```

```
baseline_summary <- read.csv("simulation_outputs/baseline_summary.csv")
summary_cv <- read.csv("simulation_outputs/summary_cv.csv")
summary_N1 <- read.csv("simulation_outputs/summary_N1.csv")
summary_ratio <- read.csv("simulation_outputs/summary_ratio.csv")
```

```
# Generate target particles
points <- tibble(
  x = runif(n_target, min = 0, max = region_side),
  y = runif(n_target, min = 0, max = region_side)
)

# Known average marker dose used in the estimator
known_marker_dose <- N1 * tablet_mean

# Generate per-tablet marker doses, then sum them
tablet_sd <- tablet_mean * tablet_cv
tablet_doses <- rnorm(N1, mean = tablet_mean, sd = tablet_sd)
tablet_doses <- pmax(round(tablet_doses), 1)

true_marker_total <- sum(tablet_doses)

# Generate marker particles
markers <- tibble(
  x = runif(true_marker_total, min = 0, max = region_side),
  y = runif(true_marker_total, min = 0, max = region_side)
)
```

Baseline analysis

```
set.seed(456)

baseline_result <- simulate_fovs_repeated(
  n_rep = 5000,
  n_target = 10000,
  N1 = 1,
  tablet_mean = 5000,
  tablet_cv = 0,
  region_side = 1,
  fov_size = 0.1,
  n_cal = 20,
  n_ext = 20,
  omega = 2
)

baseline_summary <- baseline_result$summary

baseline_summary %>%
  select(
    tablet_cv,
    mean_true_conc,
    mean_estimated_conc,
    mean_theoretical_error,
    empirical_cv,
    mean_bias,
    rmse,
    mean_effort,
  ) %>%
  knitr::kable(digits = 2, caption = "Baseline simulation results")
```

Table 1: Baseline simulation results

tablet_cv	mean_true_conc	mean_estimated_conc	mean_theoretical_error	empirical_cv	mean_bias	rmse	mean_effort
0	10000	10011.09	3.86	3.47	11.09	347.83	3079.19

Tablet CV analysis

Simulation results across tablet CV levels

```
tablet_cv_levels <- c(0, 0.01, 0.03, 0.05, 0.07, 0.09, 0.11)

results_cv <- vector("list", length(tablet_cv_levels))

for (i in seq_along(tablet_cv_levels)) {
  results_cv[[i]] <- simulate_fovs_repeated(
    n_rep = 5000,
```

```

    n_target = 10000,
    N1 = 1,
    tablet_mean = 5000,
    tablet_cv = tablet_cv_levels[i],
    region_side = 1,
    fov_size = 0.1,
    n_cal = 20,
    n_ext = 20,
    omega = 2
  )$summary
}

summary_cv <- bind_rows(results_cv)

```

```

summary_cv %>%
  select(
    tablet_cv,
    mean_true_conc,
    mean_estimated_conc,
    mean_theoretical_error,
    empirical_cv,
    mean_bias,
    rmse,
    mean_effort,
  ) %>%
  knitr::kable(digits = 2, caption = "Simulation results across tablet CV levels")

```

Table 2: Simulation results across tablet CV levels

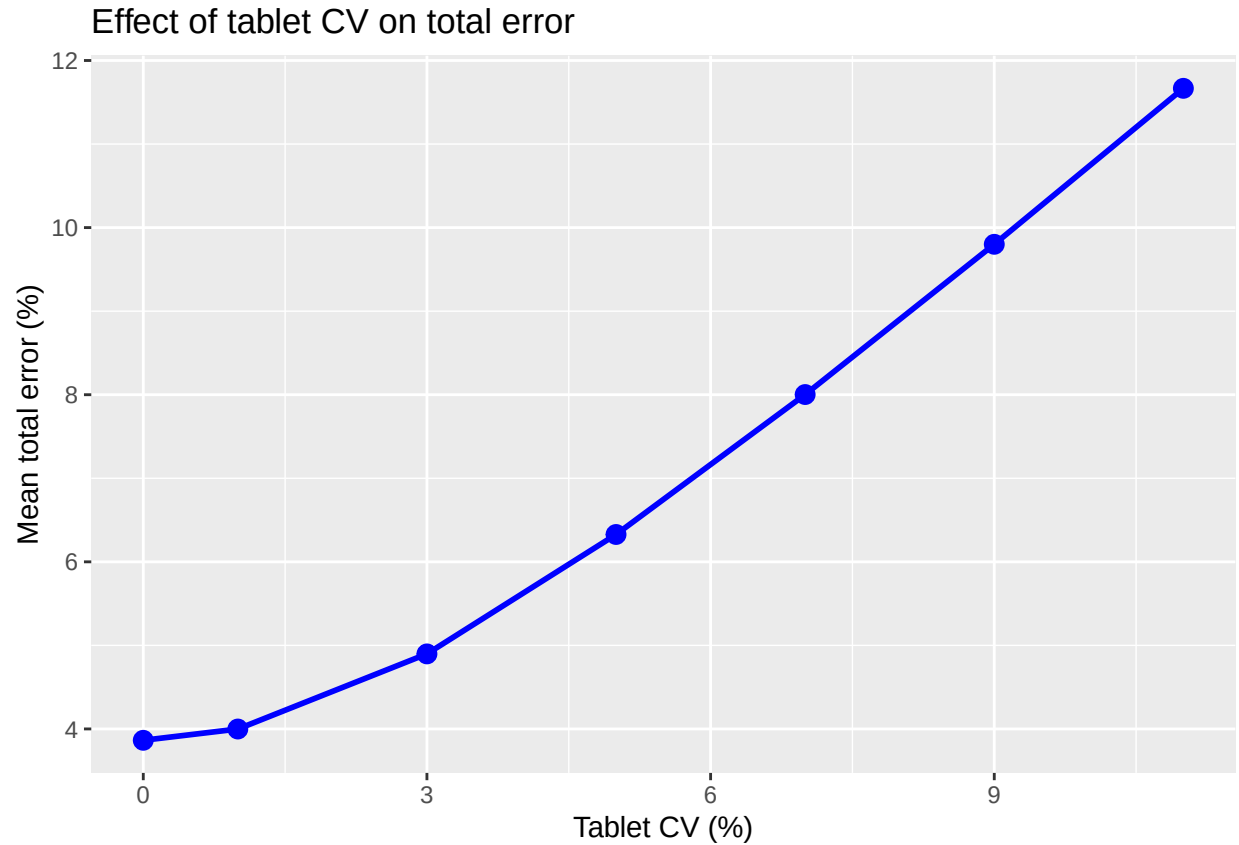
tablet_cv	mean_true_conc	mean_estimated_conc	mean_theoretical_error	empirical_cv	mean_bias	rmse	mean_effort
0.00	10000	10011.09	3.86	3.47	11.09	347.83	3079.19
0.01	10000	10003.99	4.00	3.64	3.99	364.34	3079.84
0.03	10000	10020.99	4.90	4.60	20.99	461.83	3081.06
0.05	10000	10031.62	6.33	6.14	31.62	616.96	3079.91
0.07	10000	10046.27	8.00	7.74	46.27	778.67	3080.23
0.09	10000	10074.64	9.80	9.87	74.64	997.05	3081.11
0.11	10000	10147.97	11.67	11.89	147.97	1215.13	3078.35

Plot: tablet CV and total error

```

ggplot(summary_cv, aes(x = tablet_cv * 100, y = mean_theoretical_error)) +
  geom_line(linewidth = 1, colour = "blue") +
  geom_point(size = 3, colour = "blue") +
  labs(
    title = "Effect of tablet CV on total error",
    x = "Tablet CV (%)",
    y = "Mean total error (%)"
  )

```



Error decomposition across tablet CV levels

```
summary_cv %>%
  select(
    tablet_cv,
    mean_tablet_term,
    mean_calibration_term,
    mean_marker_term,
    mean_theoretical_error
  ) %>%
  knitr::kable(digits = 2, caption = "Error decomposition across tablet CV levels")
```

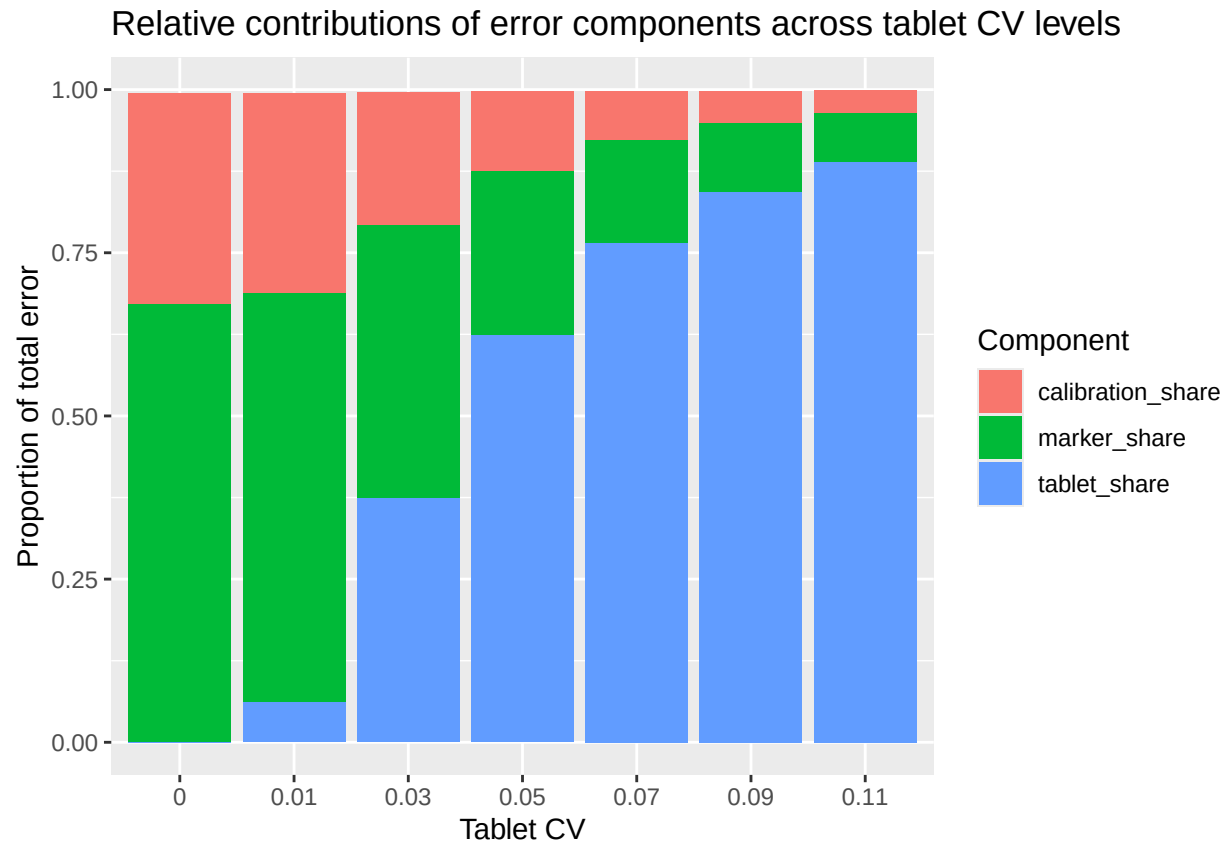
Table 3: Error decomposition across tablet CV levels

tablet_cv	mean_tablet_term	mean_calibration_term	mean_marker_term	mean_theoretical_error
0.00	0	2.20	3.16	3.86
0.01	1	2.21	3.16	4.00
0.03	3	2.21	3.16	4.90
0.05	5	2.21	3.17	6.33
0.07	7	2.21	3.17	8.00
0.09	9	2.20	3.17	9.80
0.11	11	2.20	3.18	11.67

```
summary_cv <- summary_cv %>%
  mutate(
    tablet_share = mean_tablet_term^2 / mean_theoretical_error^2,
    calibration_share = mean_calibration_term^2 / mean_theoretical_error^2,
    marker_share = mean_marker_term^2 / mean_theoretical_error^2
  )
```

```
summary_cv_share_long <- summary_cv %>%
  select(
    tablet_cv,
    tablet_share,
    calibration_share,
    marker_share
  ) %>%
  tidyr::pivot_longer(
    cols = c(tablet_share, calibration_share, marker_share),
    names_to = "component",
    values_to = "share"
  )
```

```
ggplot(summary_cv_share_long, aes(x = factor(tablet_cv), y = share, fill = component)) +
  geom_col() +
  labs(
    title = "Relative contributions of error components across tablet CV levels",
    x = "Tablet CV",
    y = "Proportion of total error",
    fill = "Component"
  )
```



N1 analysis

Simulation results across numbers of tablets

```
N1_levels <- c(1, 2, 4, 6)

total_marker_dose <- 5000
tablet_mean_levels <- total_marker_dose / N1_levels

results_N1 <- vector("list", length(N1_levels))

for (i in seq_along(N1_levels)) {
  results_N1[[i]] <- simulate_fovs_repeated(
    n_rep = 5000,
    n_target = 10000,
    N1 = N1_levels[i],
    tablet_mean = tablet_mean_levels[i],
    tablet_cv = 0.04,
    region_side = 1,
    fov_size = 0.1,
    n_cal = 20,
    n_ext = 20,
    omega = 2
  )
}
```

```

    )$summary
  }

summary_N1 <- bind_rows(results_N1)

summary_N1 %>%
  select(
    N1,
    tablet_cv,
    tablet_mean,
    mean_true_conc,
    mean_estimated_conc,
    mean_theoretical_error,
    empirical_cv,
    mean_bias,
    rmse,
    mean_effort,
  ) %>%
  knitr::kable(digits = 2, caption = "Simulation results across numbers of tablets")

```

Table 4: Simulation results across numbers of tablets

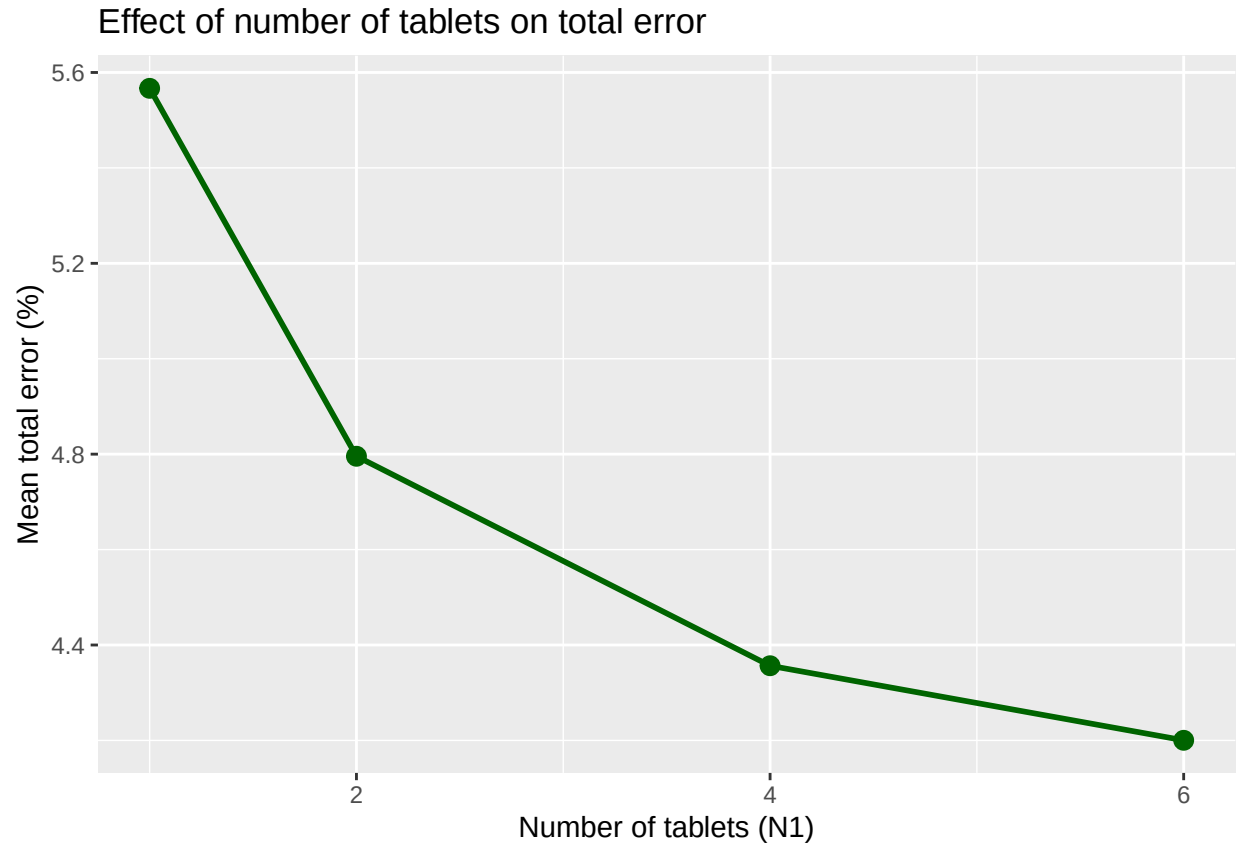
N1	tablet_cv	tablet_mean	mean_true_conc	mean_estimated_conc	mean_theoretical_error	empirical_cv	mean_bias	rmse	mean_effort
1	0.04	5000.00	10000	10032.98	5.57	5.30	32.98	532.90	3078.21
2	0.04	2500.00	10000	10016.91	4.80	4.44	16.91	445.17	3079.84
4	0.04	1250.00	10000	10014.14	4.36	3.98	14.14	398.66	3079.77
6	0.04	833.33	10000	10007.29	4.20	3.85	7.29	385.23	3079.42

Plot: number of tablets and total error

```

ggplot(summary_N1, aes(x = N1, y = mean_theoretical_error)) +
  geom_line(linewidth = 1, colour = "darkgreen") +
  geom_point(size = 3, colour = "darkgreen") +
  labs(
    title = "Effect of number of tablets on total error",
    x = "Number of tablets (N1)",
    y = "Mean total error (%)"
  )

```



Error decomposition across numbers of tablets

```
summary_N1 %>%
  select(
    N1,
    mean_tablet_term,
    mean_calibration_term,
    mean_marker_term,
    mean_theoretical_error
  ) %>%
  knitr::kable(digits = 2, caption = "Error decomposition across numbers of tablets")
```

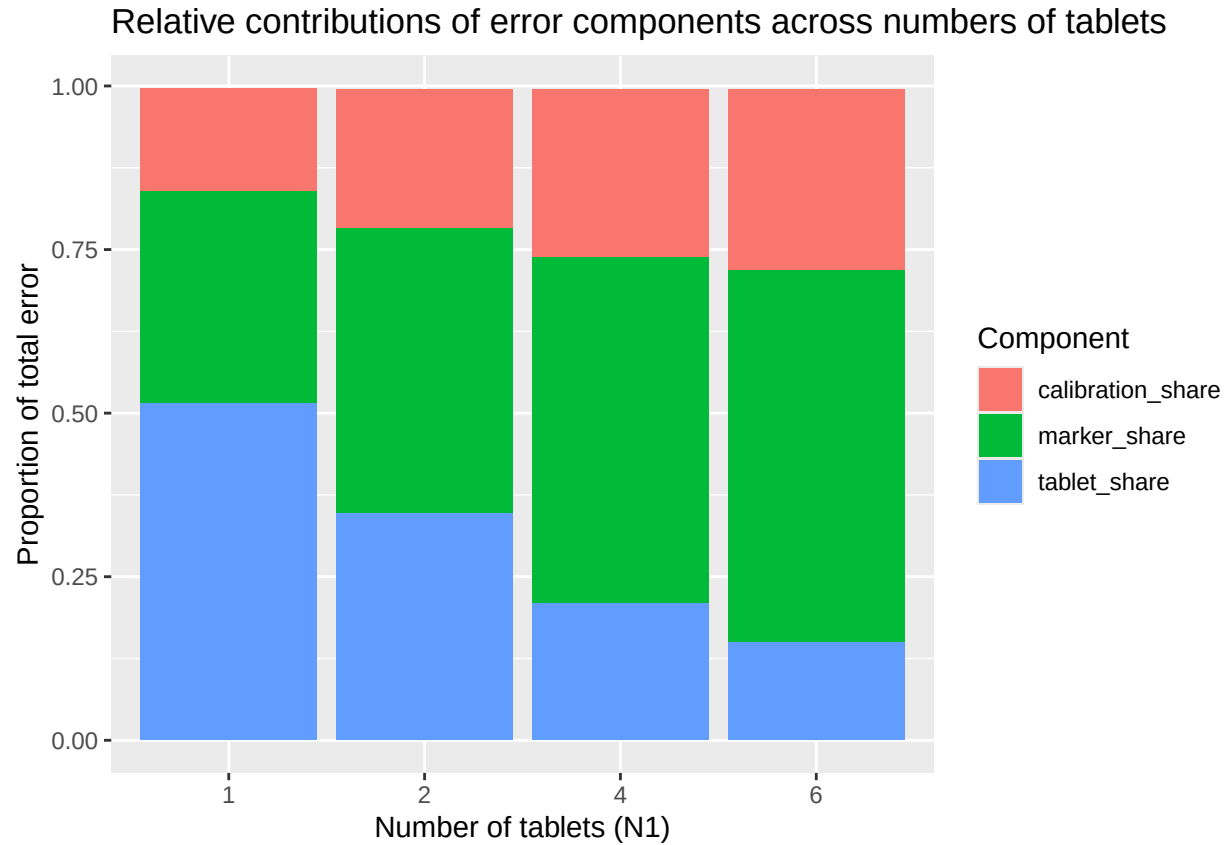
Table 5: Error decomposition across numbers of tablets

N1	mean_tablet_term	mean_calibration_term	mean_marker_term	mean_theoretical_error
1	4.00	2.20	3.17	5.57
2	2.83	2.21	3.16	4.80
4	2.00	2.21	3.16	4.36
6	1.63	2.21	3.16	4.20

```
summary_N1 <- summary_N1 %>%
  mutate(
    tablet_share = mean_tablet_term^2 / mean_theoretical_error^2,
    calibration_share = mean_calibration_term^2 / mean_theoretical_error^2,
    marker_share = mean_marker_term^2 / mean_theoretical_error^2
  )
```

```
summary_N1_share_long <- summary_N1 %>%
  select(
    N1,
    tablet_share,
    calibration_share,
    marker_share
  ) %>%
  tidyr::pivot_longer(
    cols = c(tablet_share, calibration_share, marker_share),
    names_to = "component",
    values_to = "share"
  )
```

```
ggplot(summary_N1_share_long, aes(x = factor(N1), y = share, fill = component)) +
  geom_col() +
  labs(
    title = "Relative contributions of error components across numbers of tablets",
    x = "Number of tablets (N1)",
    y = "Proportion of total error",
    fill = "Component"
  )
```



Target-to-marker ratio analysis

$$u = \frac{n_{\text{target}}}{N_1 \times \mu_{\text{tablet}}}$$

```
set.seed(456)

ratio_levels <- c(1, 2, 4, 6, 8, 10, 15)
n_target <- 10000

results_ratio <- vector("list", length(ratio_levels))

for (i in seq_along(ratio_levels)) {
  current_ratio <- ratio_levels[i]
  current_tablet_mean <- n_target / current_ratio

  results_ratio[[i]] <- simulate_fovs_repeated(
    n_rep = 5000,
    n_target = n_target,
    N1 = 1,
    tablet_mean = current_tablet_mean,
    tablet_cv = 0.04,
    region_side = 1,
    fov_size = 0.1,
  )
}
```

```

    n_cal = 20,
    n_ext = 20,
    omega = 2
  )$summary
}

summary_ratio <- bind_rows(results_ratio)
summary_ratio$ratio <- ratio_levels

```

Simulation results across target-to-marker ratio levels

```

summary_ratio %>%
  select(
    ratio,
    N1,
    tablet_cv,
    tablet_mean,
    mean_true_conc,
    mean_estimated_conc,
    mean_theoretical_error,
    empirical_cv,
    mean_bias,
    rmse,
    mean_effort,
    failure_rate
  ) %>%
  knitr::kable(digits = 2, caption = "Simulation results across target-to-marker ratio levels")

```

Table 6: Simulation results across target-to-marker ratio levels

ratio	N1	tablet_cv	tablet_mean	mean_true_conc	mean_estimated_conc	mean_theoretical_error	empirical_cv	mean_bias	rmse	mean_effort	failure_rate
1	1	0.04	10000.00	10000	10026.91	5.10	4.89	26.91	490.89	4078.43	0
2	1	0.04	5000.00	10000	10018.62	5.57	5.24	18.62	525.38	3079.36	0
4	1	0.04	2500.00	10000	10025.48	6.41	6.02	25.48	604.27	2579.89	0
6	1	0.04	1666.67	10000	10038.58	7.15	6.70	38.58	673.87	2412.98	0
8	1	0.04	1250.00	10000	10043.79	7.82	7.22	43.79	726.72	2330.57	0
10	1	0.04	1000.00	10000	10068.92	8.44	7.90	68.92	798.55	2279.91	0
15	1	0.04	666.67	10000	10090.61	9.83	9.18	90.61	930.95	2212.30	0

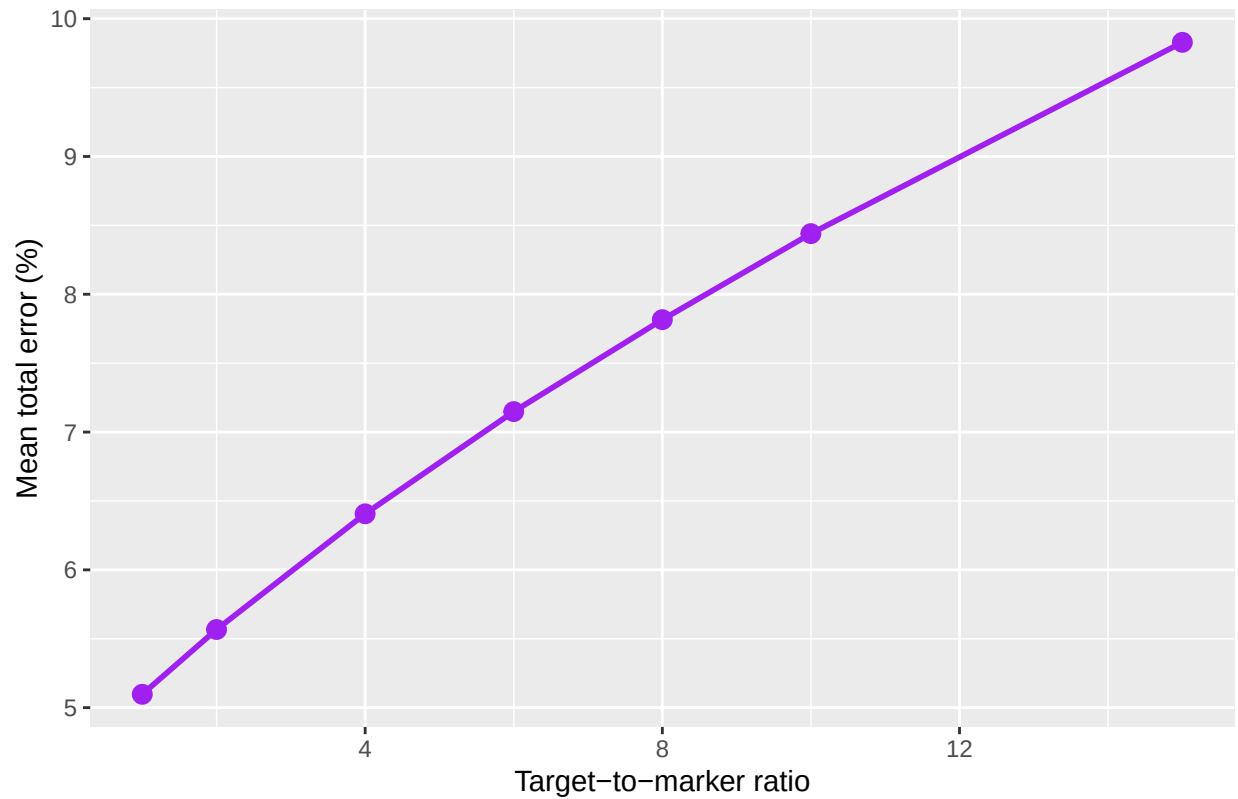
Plot: target-to-marker ratio vs total error

```

ggplot(summary_ratio, aes(x = ratio, y = mean_theoretical_error)) +
  geom_line(linewidth = 1, colour = "purple") +
  geom_point(size = 3, colour = "purple") +
  labs(
    title = "Effect of target-to-marker ratio on total error",
    x = "Target-to-marker ratio",
    y = "Mean total error (%)"
  )

```

Effect of target-to-marker ratio on total error



Error decomposition across target-to-marker ratio levels

```
summary_ratio %>%
  select(
    ratio,
    mean_tablet_term,
    mean_calibration_term,
    mean_marker_term,
    mean_theoretical_error
  ) %>%
  knitr::kable(digits = 2, caption = "Error decomposition across target-to-marker ratio levels")
```

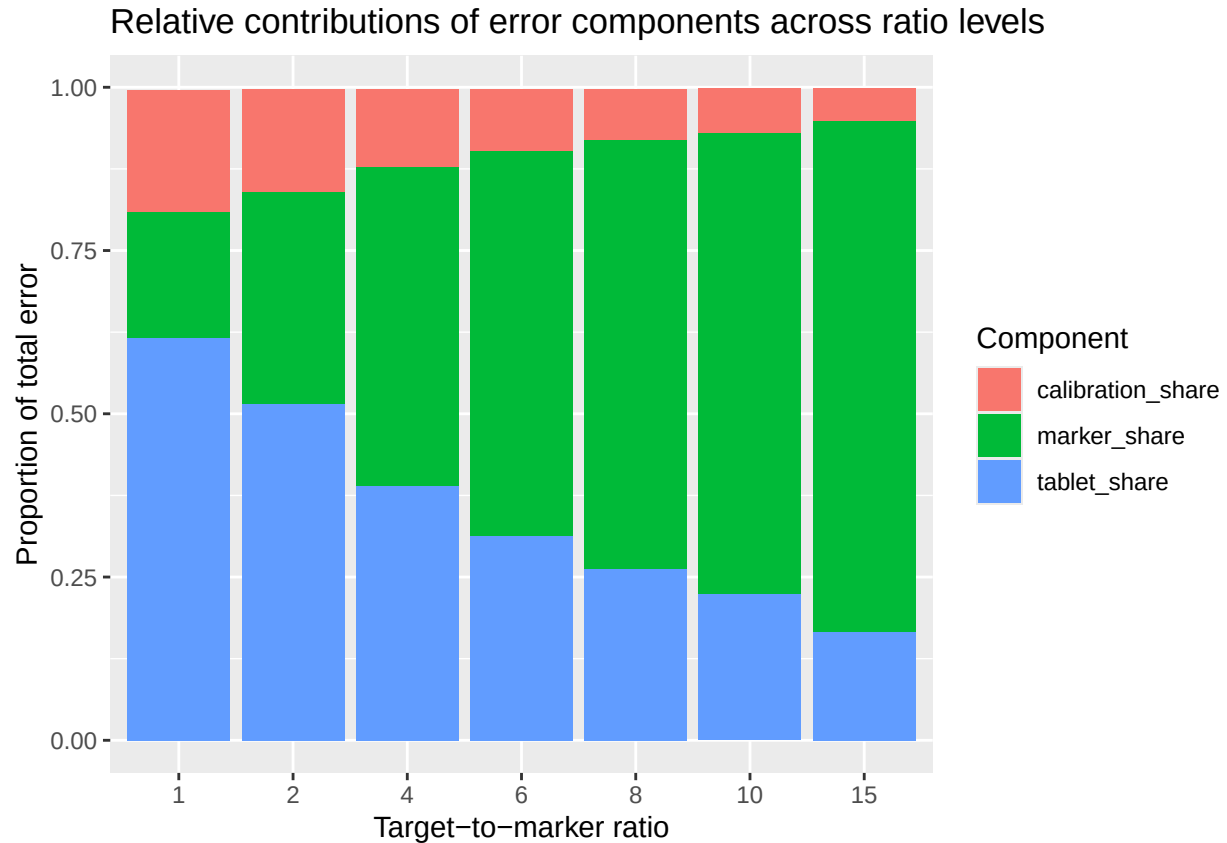
Table 7: Error decomposition across target-to-marker ratio levels

ratio	mean_tablet_term	mean_calibration_term	mean_marker_term	mean_theoretical_error
1	4	2.21	2.24	5.10
2	4	2.20	3.16	5.57
4	4	2.21	4.48	6.41
6	4	2.21	5.49	7.15
8	4	2.20	6.33	7.82
10	4	2.20	7.09	8.44
15	4	2.21	8.69	9.83

```
summary_ratio <- summary_ratio %>%
  mutate(
    tablet_share = mean_tablet_term^2 / mean_theoretical_error^2,
    calibration_share = mean_calibration_term^2 / mean_theoretical_error^2,
    marker_share = mean_marker_term^2 / mean_theoretical_error^2
  )
```

```
summary_ratio_share_long <- summary_ratio %>%
  select(
    ratio,
    tablet_share,
    calibration_share,
    marker_share
  ) %>%
  tidyr::pivot_longer(
    cols = c(tablet_share, calibration_share, marker_share),
    names_to = "component",
    values_to = "share"
  )
```

```
ggplot(summary_ratio_share_long, aes(x = factor(ratio), y = share, fill = component)) +
  geom_col() +
  labs(
    title = "Relative contributions of error components across ratio levels",
    x = "Target-to-marker ratio",
    y = "Proportion of total error",
    fill = "Component"
  )
```



Presentation slides outline

Slide 1. Title

- Project title The effect of tablet error on the precision of estimating the absolute abundance of microfossils in the FOVS method
- Name: Menghan Fu
- Student ID: a1941288
- Supervisor: Anthony Mays

Slide 2. Introduction

What is this project about?

- Specifically focuses on one part of its error structure: tablet error.
- To understand when tablet error matters and when it may be ignored.

Why do we care about fossils (Motivation)?

Slide 3. Research Questions

- Under what conditions does tablet error become an important part of total FOVS error?

- How does the importance of tablet error change under different experimental designs and sample conditions?
- If tablet error is large, are there design adjustments that can reduce its impact?

Slide 4. Background

- absolute abundance
- exotic marker method
- FOVS overview

Slide 5. Background

- FOVS error structure
- tablet error definition
- tablet CV
- research gap

Slide 6. Methods Overview

How to do the simulation? (Place a simple workflow diagram)

- Set scenario parameters
- generate fixed-count targets with random locations
- generate random per-tablet marker doses
- sum tablet doses and generate marker locations
- perform calibration and extrapolation counts
- calculate concentration and error components
- repeat simulations and summarise results

Slide 7. Methods: Scenario Design

Tablet CV scenario

- Fixed: $n_{\text{target}} = 10000$, $N1 = 1$, $\text{tablet_mean} = 5000$
- Varied: tablet_cv

N1 scenario

- Fixed: total marker dose = 5000, $\text{tablet_cv} = 0.04$, n_{target}
- Varied: $N1$
- $\text{tablet_mean} = \text{total marker dose} / N1$

Ratio scenario

- Fixed: n_{target} , $N1 = 1$, $\text{tablet_cv} = 0.04$
- Varied: tablet_mean (This changes the expected target-to-marker ratio)

Slide 8. Output Metrics

- Mean theoretical error
- empirical_cv
- RMSE
- Mean effort
- Error decomposition
- relative contribution of each error component

Slide 9. Results: Tablet CV Scenario

Figure:

- total error vs tablet CV
- relative contributions of error components across tablet CV levels

Key message:

- total error increases as tablet CV increases
- tablet_share becomes larger at higher tablet CV
- tablet error can become a dominant source of total error

Slide 10. Results: N1 Scenario

Figure:

- total error vs N1
- relative contributions across N1

Key message:

- increasing N1 reduces total error
- after revising the scenario to keep total marker dose fixed, increasing N1 mainly reduces tablet-related uncertainty
- marker uncertainty no longer improves automatically

Slide 11. Results: Ratio Scenario

Figure:

- total error vs ratio
- relative contributions across ratio

Key message:

- total error increases as the target-to-marker ratio increases
- this increase is mainly driven by the marker term
- tablet term stays roughly constant, but its relative contribution declines

Slide 12. Discussion & Conclusion

- Tablet error is not equally important under all conditions.
- Its importance changes with:
 - tablet dose variation
 - number of tablets
 - target-to-marker ratio
- Increasing the number of tablets can reduce tablet-related uncertainty.
- When tablet CV is high, tablet uncertainty becomes more important.
- When the target-to-marker ratio is high, marker counting uncertainty becomes dominant.